

BÀI THỰC HÀNH CƠ SỞ DỮ LIỆU

CÀI ĐẶT BẢNG KHOA VÀ BẢNG SINH VIÊN TRÊN SQL SERVER

Họ và tên: Nguyễn Quang Chiến

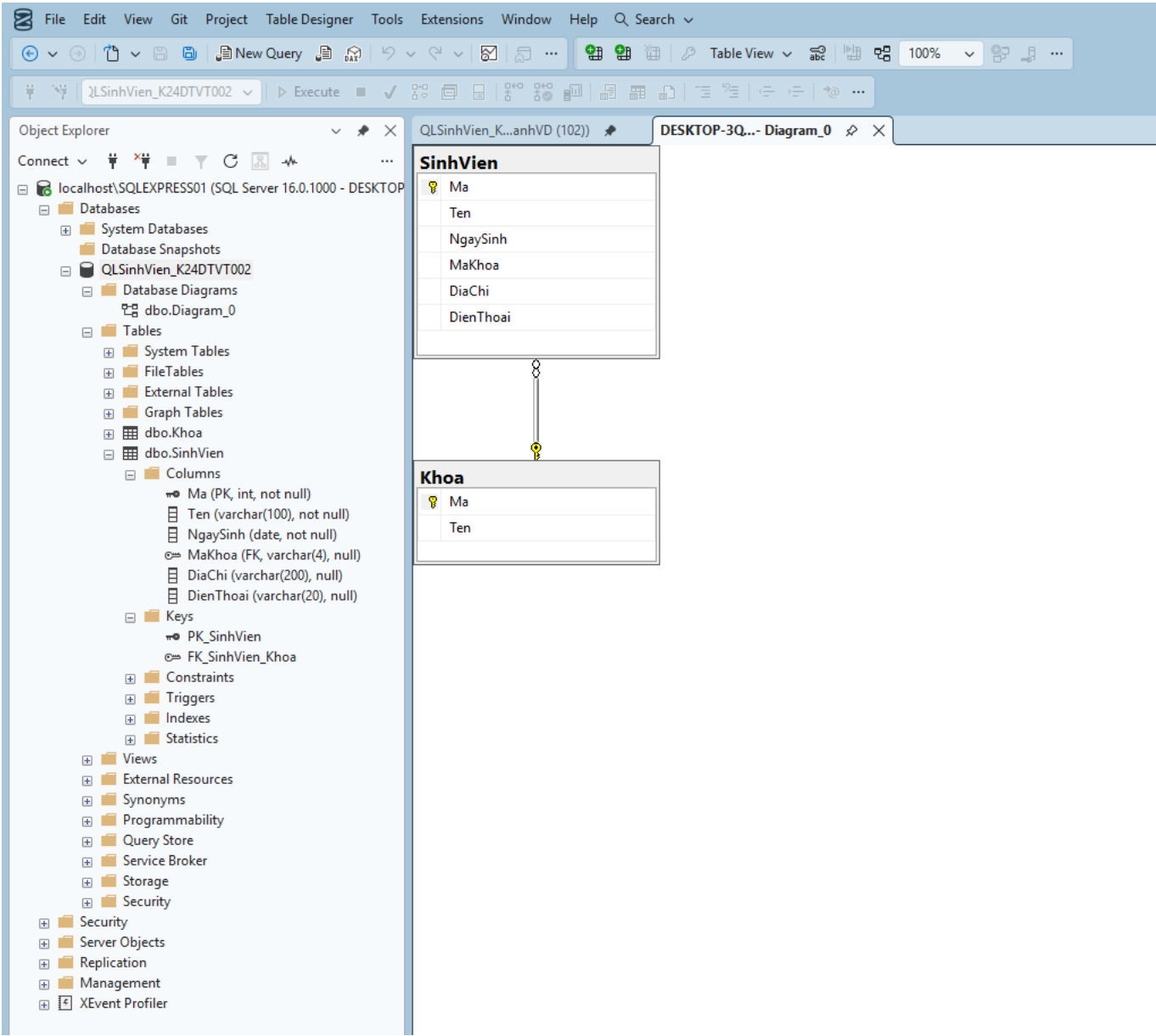
Mã sinh viên: K24DTVT002

Lớp: D24TXVT01-K

Tên CSDL:

QLSinhVien_K24DTVT002

Ảnh/lược đồ Database Diagram thể hiện đầy đủ bảng Khoa và SinhVien cùng quan hệ khóa ngoại



Ảnh cấu trúc hai bảng sau khi cài đặt

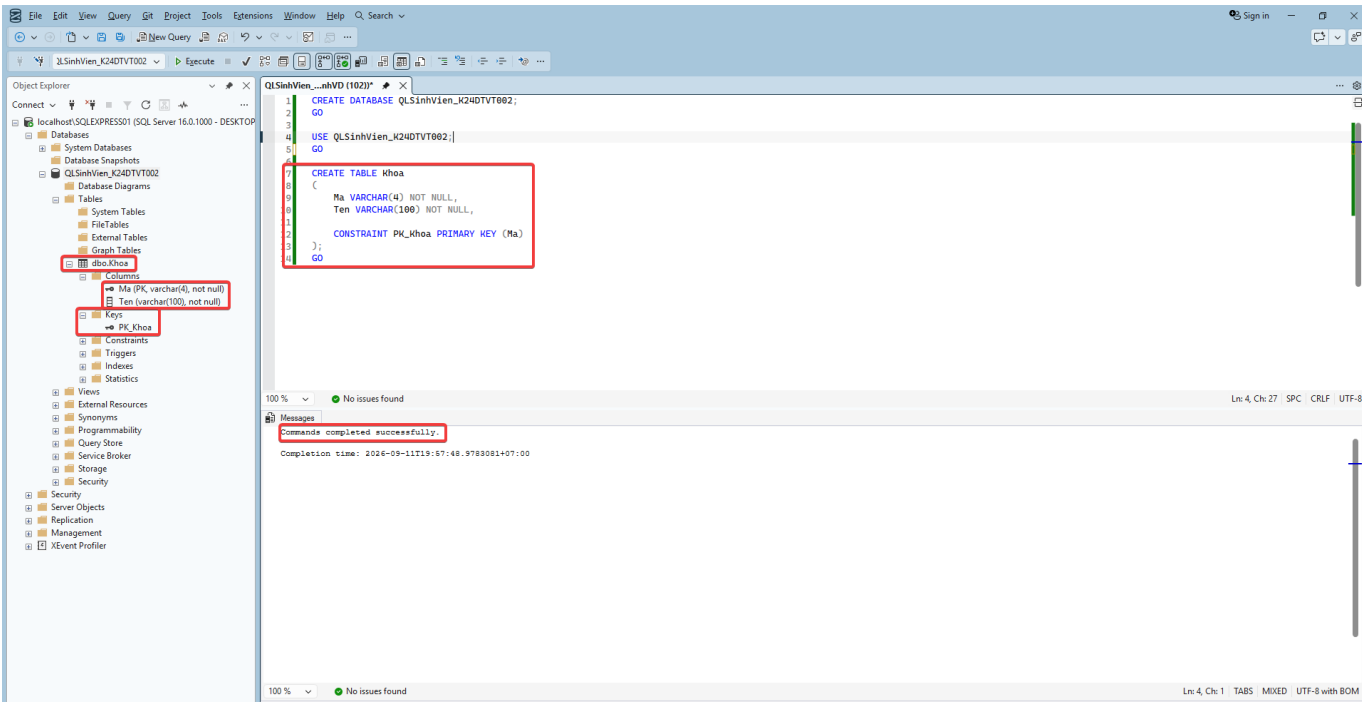
Object Explorer		QLSinhVien_K24DVT002		SQLQuery2.s...anhVD (823)*		DESKTOP-3QA... - Diagram_0	
Connect		100 %		No issues found		Ln: 4, Ch: 1 TABS LF UT	
Databases		Name		Owner		Type	
System Databases		Khoa		dbo		user table	
Database Snapshots		Created_datetime		2026-09-11 19:57:48.973			
QLSinhVien_K24DVT002		Column_name		Type		Computed	
Database Diagrams		Ma		varchar		no	
dbo.Khoa		Ten		varchar		no	
Tables		Length		4		Prec	
System Tables		Scale		100		Nullable	
FileTables		TenTrailingBlanks		no		no	
External Tables		FixedLenNullInSource		no		no	
Graph Tables		Collation		SQL_Latin1_General_CP1_CI_AS		SQL_Latin1_General_CP1_CI_AS	
Columns		Identity		Seed		Increment	
Ma (PK, varchar(4), not null)		No identity column defined.		NULL		NULL	
Ten (varchar(100), not null)		RowGuidCol		No rowguidcol column defined.			
NgaySinh (date, not null)		Data_Joined_on_filegroup		PRIMARY			
MaKhoa (FK, varchar(4), null)		index_name		index_description		index_keys	
DiaChi (varchar(200), null)		PK_Khoa		clustered, unique, primary key located on PRIMARY		Ma	
DienThoai (varchar(20), null)		constraint_type		constraint_name		delete_action	
PK_SinhVien		PRIMARY KEY (clustered)		PK_Khoa		(n/a)	
FK_SinhVien_Khoa		Table is referenced by foreign key		QLSinhVien_K24DVT002.dbo.SinhVien			
Columns		Name		Owner		Type	
Ma (PK, int, not null)		SinhVien		dbo		user table	
Ten (varchar(100), not null)		Created_datetime		2026-09-11 20:14:49.710			
NgaySinh (date, not null)		Column_name		Type		Computed	
MaKhoa (FK, varchar(4), null)		Ma		int		no	
DiaChi (varchar(200), null)		Ten		varchar		no	
DienThoai (varchar(20), null)		NgaySinh		date		no	
PK_SinhVien		MaKhoa		varchar		no	
FK_SinhVien_Khoa		DiaChi		varchar		no	
Columns		DienThoai		varchar		no	
Ma (PK, int, not null)		Identity		Seed		Increment	
Ten (varchar(100), not null)		No identity column defined.		NULL		NULL	
NgaySinh (date, not null)		RowGuidCol		No rowguidcol column defined.			
MaKhoa (FK, varchar(4), null)		Data_Joined_on_filegroup		PRIMARY			
DiaChi (varchar(200), null)		index_name		index_description		index_keys	
DienThoai (varchar(20), null)		PK_SinhVien		clustered, unique, primary key located on PRIMARY		Ma	
PK_SinhVien		constraint_type		constraint_name		delete_action	
FK_SinhVien_Khoa		FOREIGN KEY		FK_SinhVien_Khoa		No Action	
Columns		status_enabled		status_for_replication		constraint_keys	
Ma (PK, int, not null)		Enabled		Is_For_Replication		MaKhoa	
Ten (varchar(100), not null)		REFERENCES QLSinhVien_K24DVT002.dbo.Khoa (Ma)					

3.1. Tạo cơ sở dữ liệu

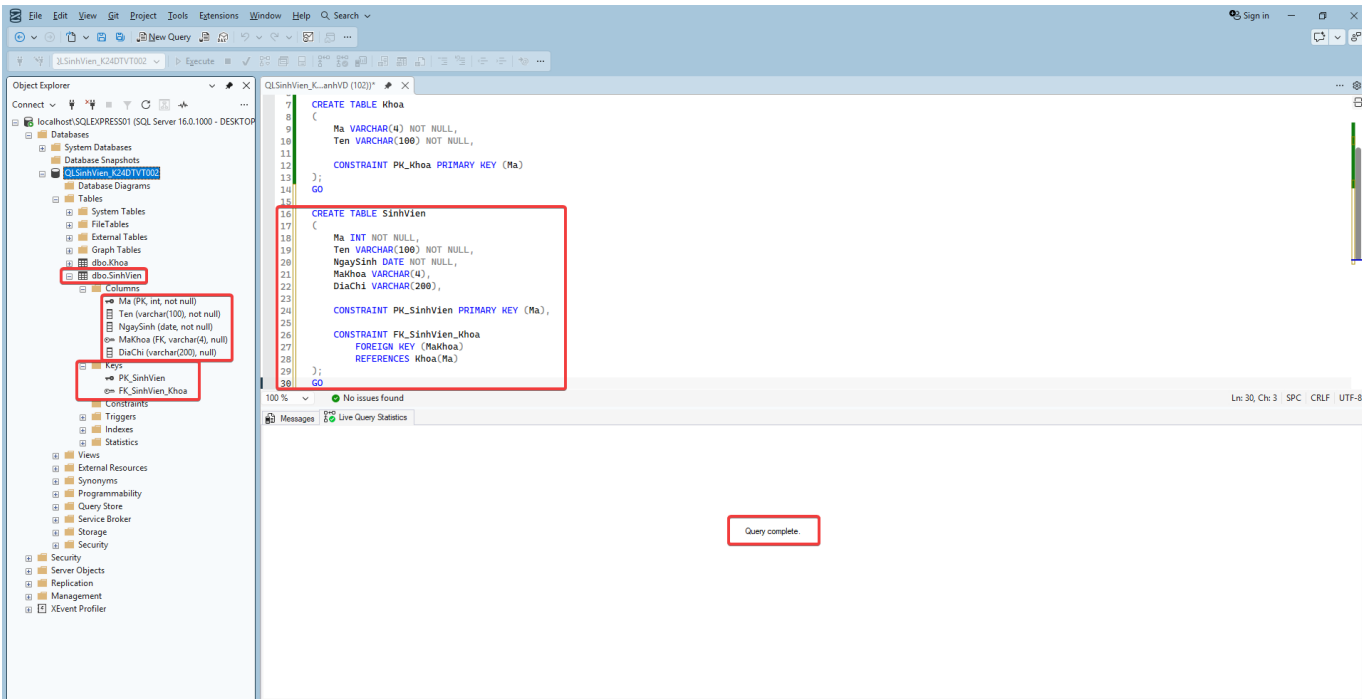
Object Explorer		QLSinhVien...anhVD (102)*		SQLQuery2.s...anhVD (823)*		DESKTOP-3QA... - Diagram_0	
Connect		100 %		No issues found		Ln: 5, Ch: 3 SPC CRLF UT	
Databases		Name		Owner		Type	
System Databases		Khoa		dbo		user table	
Database Snapshots		Created_datetime		2026-09-11 20:14:49.710			
QLSinhVien_K24DVT002		Column_name		Type		Computed	
Database Diagrams		Ma		int		no	
dbo.Khoa		Ten		varchar		no	
Tables		NgaySinh		date		no	
System Tables		MaKhoa		varchar		no	
FileTables		DiaChi		varchar		no	
External Tables		DienThoai		varchar		no	
Graph Tables		Identity		Seed		Increment	
Columns		No identity column defined.		NULL		NULL	
Ma (PK, varchar(4), not null)		RowGuidCol		No rowguidcol column defined.			
Ten (varchar(100), not null)		Data_Joined_on_filegroup		PRIMARY			
NgaySinh (date, not null)		index_name		index_description		index_keys	
MaKhoa (FK, varchar(4), null)		PK_SinhVien		clustered, unique, primary key located on PRIMARY		Ma	
DiaChi (varchar(200), null)		constraint_type		constraint_name		delete_action	
DienThoai (varchar(20), null)		FOREIGN KEY		FK_SinhVien_Khoa		No Action	
PK_SinhVien		status_enabled		status_for_replication		constraint_keys	
FK_SinhVien_Khoa		Enabled		Is_For_Replication		MaKhoa	
Columns		REFERENCES QLSinhVien_K24DVT002.dbo.Khoa (Ma)					

Messages		Commands completed successfully.		Completion time: 2026-09-11T19:57:48.9783081+07:00	
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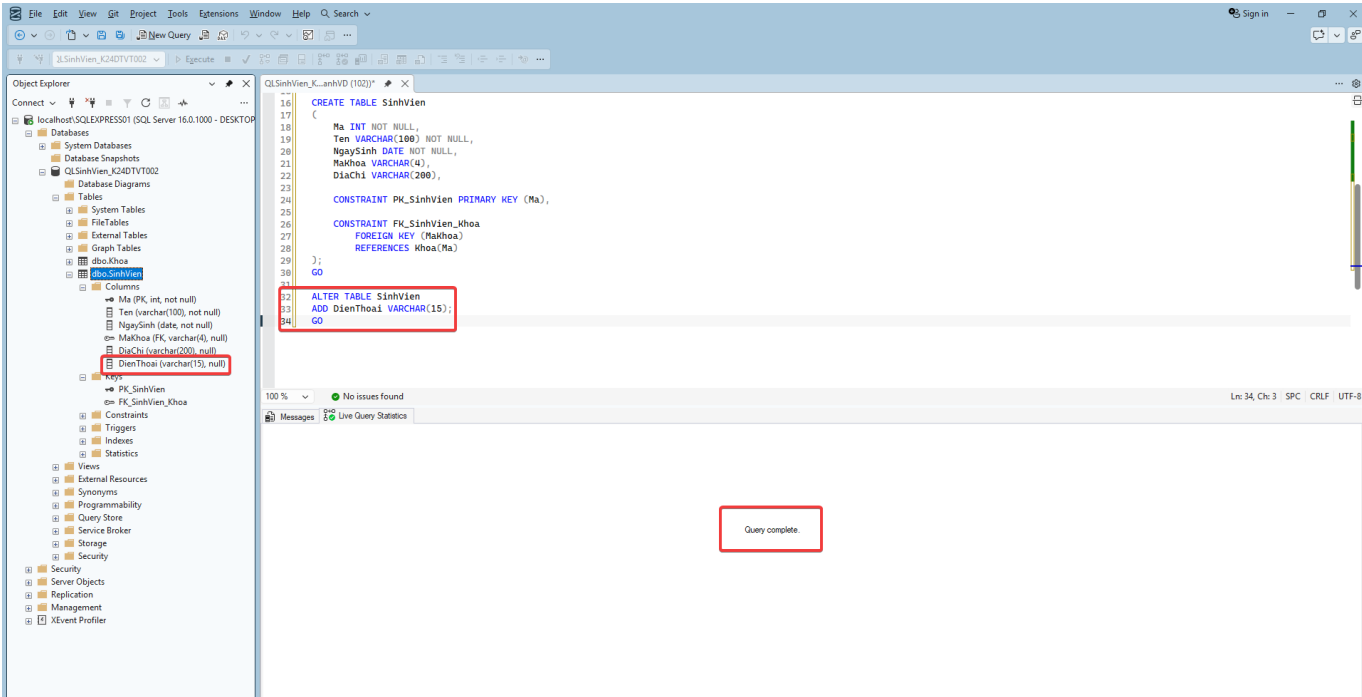
3.2. Tạo bảng Khoa



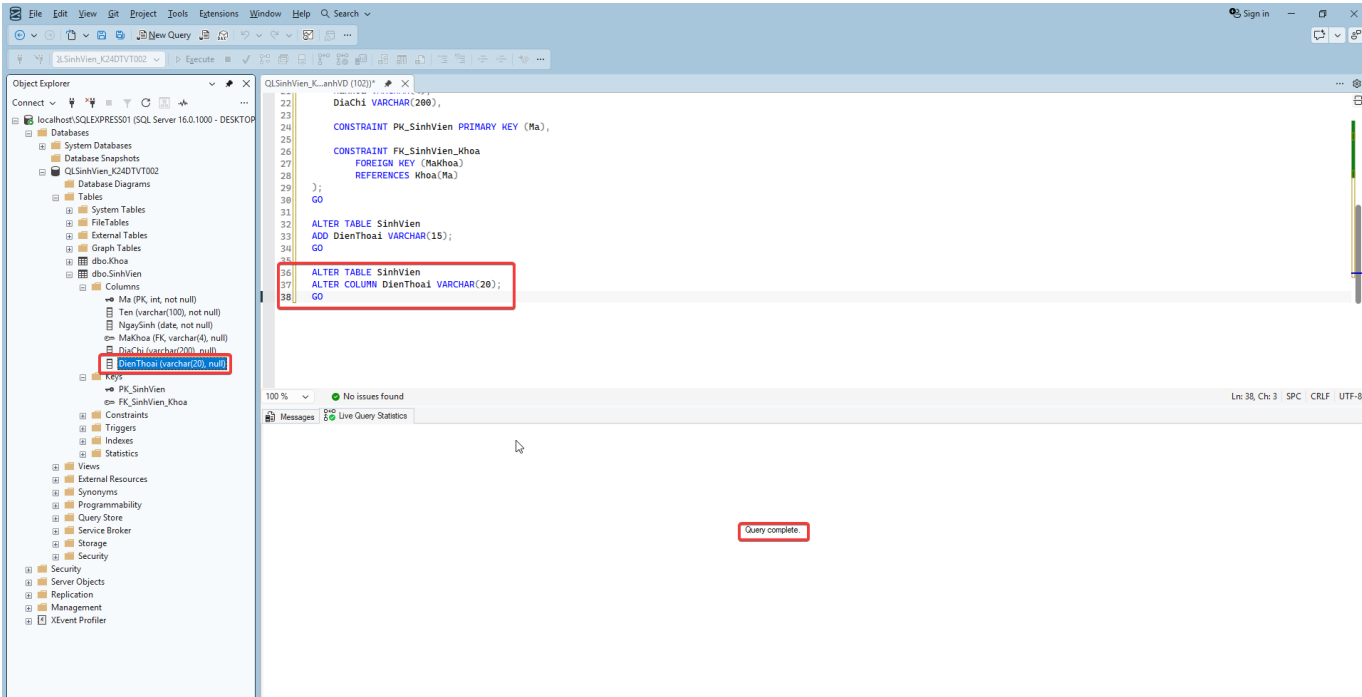
3.3. Tạo bảng SinhVien



3.4. Bổ sung cột DienThoai



3.5. Mở rộng cột DienThoai



3.6. Nhập dữ liệu mẫu

Object Explorer

Connect

localhost\SQLEXPRESS01 (SQL Server 16.0.1000 - DESKTOP)

Databases

System Databases

Database Snapshots

QLSinhVien_K24DVTY002

Database Diagrams

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.Khoa

dbo.SinhVien

Columns

Ma (PK, int, not null)

Ten (varchar(100), not null)

NgaySinh (date, not null)

MaKhoa (FK, varchar(4), null)

DiaChi (varchar(200), null)

DienThoai (varchar(20), null)

Keys

PK_SinhVien

FK_SinhVien_Khoa

Constraints

Triggers

Indexes

Statistics

Views

External Resources

Synonyms

Programmability

Query Store

Service Broker

Storage

Security

Server Objects

Replication

Management

XEvent Profiler

QLSinhVien...nhVD (102)%

REFERENCES Khoa(Ma)

);

GO

ALTER TABLE SinhVien

ADD DienThoai VARCHAR(15);

GO

ALTER TABLE SinhVien

ALTER COLUMN DienThoai VARCHAR(20);

GO

INSERT INTO Khoa (Ma, Ten)

VALUES

('CNTT', 'Công nghệ thông tin');

('QTKD', 'Quản trị kinh doanh');

('NNA', 'Ngôn ngữ Anh');

GO

SELECT * FROM Khoa;

100 %

No issues found

Ln: 47, Ch: 20 SPC CRLF UTF-8

Results

Messages

Live Query Statistics

Execution plan

	Ma	Ten
1	CNTT	Công nghệ thông tin
2	NNA	Ngôn ngữ Anh
3	QTKD	Quản trị kinh doanh

Rows	Executes	Stmt Text	StmtId	Modeld	Parent	PhysicalOp	LogicalOp	Argument	DefinedValues	EstimateRows	EstimateIO	EstimateCPU
1	3	1	SELECT * FROM Khoa	1	1	0	NULL	NULL		3	NULL	NULL
2	3	1	I-Clustered Index Scan OBJECT [QLSinhVien_K24...	1	2	1	Clustered Index Scan	Clustered Index Scan	OBJECT [QLSinhVien_K24DVTY002] [dbo] [Khoa] [Ma], [QLS...	3	0.003125	0.000

Object Explorer

Connect

localhost\SQLEXPRESS01 (SQL Server 16.0.1000 - DESKTOP)

Databases

System Databases

Database Snapshots

QLSinhVien_K24DVTY002

Database Diagrams

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.Khoa

dbo.SinhVien

Columns

Ma (PK, int, not null)

Ten (varchar(100), not null)

NgaySinh (date, not null)

MaKhoa (FK, varchar(4), null)

DiaChi (varchar(200), null)

DienThoai (varchar(20), null)

Keys

PK_SinhVien

FK_SinhVien_Khoa

Constraints

Triggers

Indexes

Statistics

Views

External Resources

Synonyms

Programmability

Query Store

Service Broker

Storage

Security

Server Objects

Replication

Management

XEvent Profiler

QLSinhVien...nhVD (102)%

SELECT * FROM Khoa;

GO

INSERT INTO SinhVien

(Ma, Ten, NgaySinh, MaKhoa, DiaChi, DienThoai)

VALUES

(1, 'Nguyen Van An', '2004-01-15', 'CNTT', 'Ha Noi', '0912345678');

(2, 'Tran Minh Duc', '2004-03-20', 'CNTT', 'Hai Phong', '0987654321');

(3, 'Le Hoang Nam', '2003-11-05', 'CNTT', 'Nam Dinh', '0901234567');

(4, 'Phan Quang Huy', '2004-06-12', 'CNTT', 'Ha Noi', '0934567890');

(5, 'Do Thu Ha', '2004-02-14', 'QTKD', 'Ha Noi', '0961234567');

(6, 'Bui Ngoc Mai', '2003-09-25', 'QTKD', 'Bac Ninh', '0972345678');

(7, 'Hoang Gia Bao', '2004-07-08', 'QTKD', 'Hai Duong', '0943456789');

(8, 'Nguyen Minh Anh', '2004-05-17', 'NNA', 'Ha Noi', '0924567890');

(9, 'Tran Thu Trang', '2003-12-01', 'NNA', 'Hung Yen', '0915678901');

(10, 'Le Khanh Linh', '2004-08-23', 'NNA', 'Ha Nam', '0986789012');

GO

SELECT * FROM SinhVien;

100 %

1 0

Ln: 66, Ch: 1 (23 chars, 1 lines) SPC CRLF UTF-8

Results

Messages

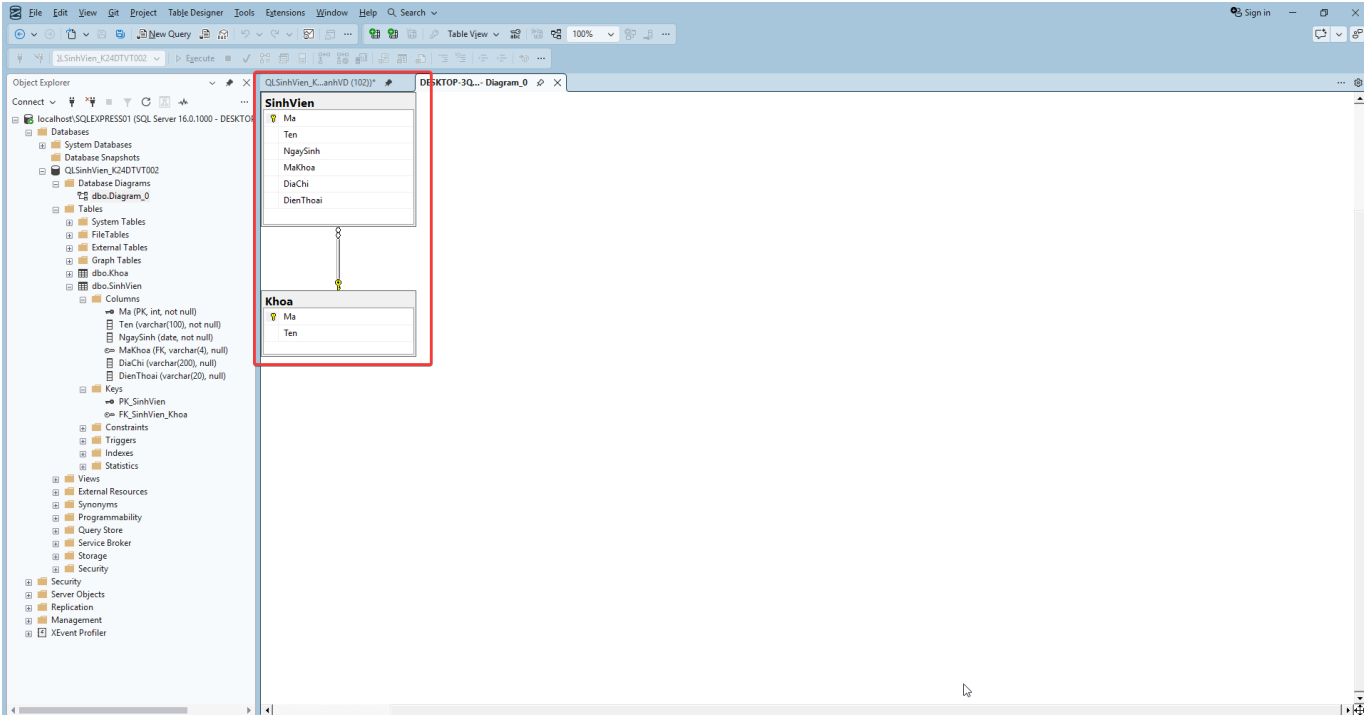
Live Query Statistics

Execution plan

	Ma	Ten	NgaySinh	MaKhoa	DiaChi	DienThoai
1	1	Nguyen Van An	2004-01-15	CNTT	Ha Noi	0912345678
2	2	Tran Minh Duc	2004-03-20	CNTT	Hai Phong	0987654321
3	3	Le Hoang Nam	2003-11-05	CNTT	Nam Dinh	0901234567
4	4	Phan Quang Huy	2004-06-12	CNTT	Ha Noi	0934567890
5	5	Do Thu Ha	2004-02-14	QTKD	Ha Noi	0961234567
6	6	Bui Ngoc Mai	2003-09-25	QTKD	Bac Ninh	0972345678
7	7	Hoang Gia Bao	2004-07-08	QTKD	Hai Duong	0943456789
8	8	Nguyen Minh Anh	2004-05-17	NNA	Ha Noi	0924567890
9	9	Tran Thu Trang	2003-12-01	NNA	Hung Yen	0915678901
10	10	Le Khanh Linh	2004-08-23	NNA	Ha Nam	0986789012

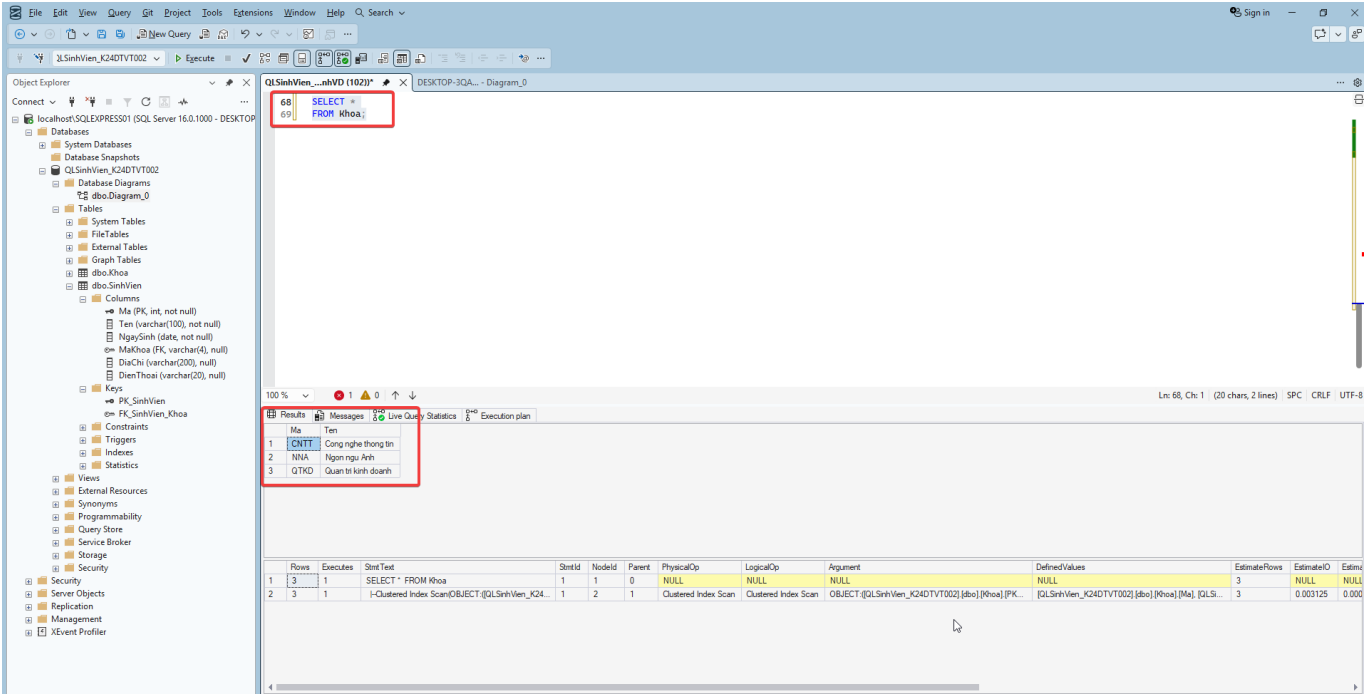
Rows	Executes	Stmt Text	StmtId	Modeld	Parent	PhysicalOp	LogicalOp	Argument	DefinedValues	EstimateRows	EstimateIO	EstimateCPU
1	10	1	SELECT * FROM SinhVien	1	1	0	NULL	NULL		10	NULL	NULL
2	10	1	I-Clustered Index Scan OBJECT [QLSinhVien_K24...	1	2	1	Clustered Index Scan	Clustered Index Scan	OBJECT [QLSinhVien_K24DVTY002] [dbo] [SinhVien...]	10	0.003125	0.00016

3.7. Kiểm tra mối quan hệ



3.8. Thực hiện truy vấn kiểm tra

Câu 1 — Hiển thị toàn bộ danh sách khoa



Câu 2 — Hiển thị toàn bộ danh sách sinh viên

Object Explorer

Connect

localhost\SQLEXPRESS01 (SQL Server 16.0.1000 - DESKTOP)

Databases

System Databases

Database Snapshots

QLSinhVien_K24DVTY1002

Database Diagrams

dbo.Diagram_0

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.Khoa

dbo.SinhVien

Columns

Ma (PK, int, not null)

Ten (varchar(100), not null)

NgaySinh (date, not null)

MaKhoa (FK, varchar(4), null)

DiaChi (varchar(200), null)

DienThoai (varchar(20), null)

Keys

PK_SinhVien

FK_SinhVien_Khoa

Constraints

Triggers

Indexes

Statistics

Views

External Resources

Synonyms

Programmability

Query Store

Service Broker

Storage

Security

Server Objects

Replication

Management

XEvent Profiler

QLSinhVien...nhVD (102)*

DESKTOP-3QA... - Diagram_0

```
68 SELECT *
69 FROM Khoa;
70
71 SELECT *
72 FROM SinhVien;
```

Results

Messages

Live Query Statistics

Execution plan

	Ma	Ten	NgaySinh	MaKhoa	DiaChi	DienThoai
1	1	Nguyen Van An	2004-01-15	CNTT	Ha Noi	0912345678
2	2	Tran Minh Duc	2004-03-20	CNTT	Ha Phong	0987654321
3	3	Le Huang Nam	2003-11-05	CNTT	Nam Dinh	0901234567
4	4	Phan Quang Huy	2004-06-12	CNTT	Ha Noi	0934567890
5	5	Do Thu Ha	2004-02-14	QTKD	Ha Noi	0961234567
6	6	Bui Ngoc Mai	2003-09-25	QTKD	Bac Ninh	0972345678
7	7	Hoang Gia Bao	2004-07-08	QTKD	Hai Duong	0943456789
8	8	Nguyen Minh Anh	2004-05-17	NNA	Ha Noi	0924567890
9	9	Tran Thu Trang	2003-12-01	NNA	Hung Yen	0915678901
10	10	Le Khanh Linh	2004-08-23	NNA	Ha Nam	0986789012

Rows	Executes	Stmt Text	StmtId	NodeId	Parent	PhysicalOp	LogicalOp	Argument	DefinedValues	EstimateRows	EstimateIO	EstimateCost
1	10	1	SELECT * FROM SinhVien	1	1	0	NULL	NULL	NULL	10	NULL	NULL
2	10	1	-Clustered Index Scan(OBJECT @[QLSinhVien_K24...	1	2	1	Clustered Index Scan	Clustered Index Scan	OBJECT @[QLSinhVien_K24DVTY1002] [dbo].[SinhVien...	10	0.003125	0.00016

Câu 3 — Liệt kê Ma, Ten, NgaySinh, DiaChi, DienThoai

Object Explorer

Connect

localhost\SQLEXPRESS01 (SQL Server 16.0.1000 - DESKTOP)

Databases

System Databases

Database Snapshots

QLSinhVien_K24DVTY1002

Database Diagrams

dbo.Diagram_0

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.Khoa

dbo.SinhVien

Columns

Ma (PK, int, not null)

Ten (varchar(100), not null)

NgaySinh (date, not null)

MaKhoa (FK, varchar(4), null)

DiaChi (varchar(200), null)

DienThoai (varchar(20), null)

Keys

PK_SinhVien

FK_SinhVien_Khoa

Constraints

Triggers

Indexes

Statistics

Views

External Resources

Synonyms

Programmability

Query Store

Service Broker

Storage

Security

Server Objects

Replication

Management

XEvent Profiler

QLSinhVien...nhVD (102)*

DESKTOP-3QA... - Diagram_0

```
68 SELECT *
69 FROM Khoa;
70
71 SELECT *
72 FROM SinhVien;
73
74 SELECT
75 Ma,
76 Ten,
77 NgaySinh,
78 DiaChi,
79 DienThoai
80 FROM SinhVien;
```

Results

Messages

Live Query Statistics

Execution plan

	Ma	Ten	NgaySinh	DiaChi	DienThoai
1	1	Nguyen Van An	2004-01-15	Ha Noi	0912345678
2	2	Tran Minh Duc	2004-03-20	Ha Phong	0987654321
3	3	Le Huang Nam	2003-11-05	Nam Dinh	0901234567
4	4	Phan Quang Huy	2004-06-12	Ha Noi	0934567890
5	5	Do Thu Ha	2004-02-14	Ha Noi	0961234567
6	6	Bui Ngoc Mai	2003-09-25	Bac Ninh	0972345678
7	7	Hoang Gia Bao	2004-07-08	Hai Duong	0943456789
8	8	Nguyen Minh Anh	2004-05-17	Ha Noi	0924567890
9	9	Tran Thu Trang	2003-12-01	Hung Yen	0915678901
10	10	Le Khanh Linh	2004-08-23	Ha Nam	0986789012

Rows	Executes	Stmt Text	StmtId	NodeId	Parent	PhysicalOp	LogicalOp	Argument	DefinedValues	EstimateRows	EstimateIO	EstimateCost
1	10	1	SELECT Ma, Ten, NgaySinh, DiaChi	1	1	0	NULL	NULL	NULL	10	NULL	NULL
2	10	1	-Clustered Index Scan(OBJECT @[QLSinhVien_K24...	1	2	1	Clustered Index Scan	Clustered Index Scan	OBJECT @[QLSinhVien_K24DVTY1002] [dbo].[SinhVien...	10	0.003125	0.00016

Câu 4 — Sinh viên kèm tên khoa bằng JOIN

The screenshot displays the Microsoft SQL Server Enterprise Edition interface. The top menu bar includes File, Edit, Query, Get, Project, Tools, Window, and Help. The left pane shows the Object Explorer with a tree view of the database structure, including Databases, System Databases, Database Snapshots, and Database Diagrams. The central pane shows a query window with a SELECT statement. The bottom pane shows the results of the query, which is a table with 10 rows of student data. The bottom-most pane shows the execution plan, which includes a clustered index scan on the SinhVien table.

Query Text:

```

SELECT
    Ma,
    Ten,
    NgaySinh,
    DiaChi,
    DienThoai
FROM SinhVien;

SELECT
    sv.Ma,
    sv.Ten AS TenSinhVien,
    sv.NgaySinh,
    sv.DiaChi,
    sv.DienThoai,
    k.Ten AS TenKhoa
FROM SinhVien sv
INNER JOIN Khoa k
    ON sv.MaKhoa = k.Ma;
  
```

Results Table:

	Ma	TenSinhVien	NgaySinh	DiaChi	DienThoai	TenKhoa
1	1	Nguyen Van An	2004-01-15	Ha Noi	0912245678	Cong nghe thong tin
2	2	Tran Minh Duc	2004-03-20	Hai Phong	0987654321	Cong nghe thong tin
3	3	Le Hoang Nam	2003-11-05	Nam Dinh	0901234567	Cong nghe thong tin
4	4	Pham Quang Huy	2004-06-12	Ha Noi	0934567890	Cong nghe thong tin
5	5	Do Thu Ha	2004-02-14	Ha Noi	0961234567	Quan tri kinh doanh
6	6	Bui Ngoc Mai	2003-09-25	Bac Ninh	0972345678	Quan tri kinh doanh
7	7	Hoang Ha Bao	2004-07-08	Hai Duong	0943456789	Quan tri kinh doanh
8	8	Nguyen Minh Anh	2004-05-17	Ha Noi	0904567890	Ngon ngu Anh
9	9	Tran Thu Trang	2003-12-01	Hung Yen	0915678901	Ngon ngu Anh
10	10	Le Khanh Linh	2004-08-23	Ha Nam	0986789012	Ngon ngu Anh

Execution Plan:

Rows	Executes	StmtText	StmtId	NodeId	Parent	PhysicalOp	LogicalOp	Argument	DefinedValues	EstimateRows	EstimateIO	EstimateCPU
1	10	1	SELECT sv.Ma, sv.Ten AS TenSinhVien, k...	1	0	NULL	NULL		NULL	10	NULL	4.18E-05
2	10	1	I-Nested Loop (inner join, OUTER REFERENCES (...))	1	2	1	1	Inner join	OUTER REFERENCES ([sv].[MaKhoa])	NULL	0	4.18E-05
3	10	1	I-Clustered Index Scan/OBJECT ([QLSinhVien_K_...])	1	3	2	Clustered Index Scan	Clustered Index Scan	OBJECT ([QLSinhVien_K24DVTY002].[dbo].[SinhVien].[...])	10	0.003125	0.000168
4	10	10	I-Clustered Index Seek/OBJECT ([QLSinhVien_K_...])	1	4	2	Clustered Index Seek	Clustered Index Seek	OBJECT ([QLSinhVien_K24DVTY002].[dbo].[PK_...])	1	0.003125	0.0001581

Câu 5 — Sinh viên thuộc một khoa

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'QLSinHVien_K240TVT002'. The 'Tables' folder is expanded, showing 'dbo.SinhVien' and 'dbo.MaKhóa'. The 'Columns' folder for 'dbo.SinhVien' is expanded, showing columns like 'Ma', 'Ten', 'NgaySinh', 'DiaChi', 'DienThoai', and 'TenKhoa'. The 'Keys' folder is also expanded, showing the primary key 'PK_SinhVien' on the 'Ma' column.

The main window displays a SQL query in the 'Query Editor' (SQL: SinHVien_K240TVT002):

```
SELECT
    sv.Ma,
    sv.Ten,
    sv.NgaySinh,
    sv.DiaChi,
    sv.DienThoai,
    k.Ten AS TenKhoa
FROM SinHVien sv
INNER JOIN Khoa k
ON sv.MaKhoa = k.Ma
WHERE sv.MaKhoa = 'CNTT';
```

The 'Results' tab shows the query results:

	Ma	Ten	NgaySinh	DiaChi	DienThoai	TenKhoa
1	1	Nguyen Van An	2004-01-15	Ha Noi	0912345678	Cong nghe thong tin
2	2	Tran Minh Duc	2004-03-20	Ha Phong	0907654321	Cong nghe thong tin
3	3	Le Huong Nam	2003-11-05	Nam Dinh	0901234567	Cong nghe thong tin
4	4	Pham Quang Huy	2004-06-12	Ha Noi	0934567890	Cong nghe thong tin

The 'Execution plan' tab shows the query execution plan:

Rows	Executes	StmtText	StmtID	NodetID	Parent	PhysicalOp	LogicalOp	Argument	DefinedValues	EstimateRows	EstimateIO	EstimateCPU	...
1	4	1	SELECT sv.Ma, sv.Ten, sv.NgaySinh, ...	1	1	0	NULL	NULL	NULL	4	NULL	NULL	...
2	4	1	1-Nested Loops (Inner Join)	1	2	1	Nested Loops	Inner Join	NULL	4	0	1.672E-05	...
3	1	1	1-Clustered Index Seek: OBJECT [QLSinHVien_K240TVT002].[dbo].[DienThoai]	1	3	2	Clustered Index Seek	Clustered Index Seek	OBJECT [QLSinHVien_K240TVT002].[dbo].[DienThoai]	1	0.003125	0.0001581	...
4	4	1	1-Clustered Index Scan: OBJECT [QLSinHVien_K240TVT002].[dbo].[SinHVien]	1	4	2	Clustered Index Scan	Clustered Index Scan	OBJECT [QLSinHVien_K240TVT002].[dbo].[SinHVien]	4	0.003125	0.000168	...

Câu 6 — Đếm số sinh viên từng khoa

Object Explorer

Connect

localhost\SQLEXPRESS01 (SQL Server 16.0.1000 - DESKTOP)

Databases

System Databases

Database Snapshots

QLSinhVien_K240TVT002

Database Diagrams

dbo.Diagram_0

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.Khoa

dbo.SinhVien

Columns

Ma (PK int, not null)

Ten (varchar(100), not null)

NgaySinh (date, not null)

MaKhoa (FK, varchar(4), null)

DiaChi (varchar(200), null)

DienThoai (varchar(20), null)

Keys

PK_SinhVien

FK_SinhVien_Khoa

Constraints

Triggers

Indexes

Statistics

Views

External Resources

Synonyms

Programmability

Query Store

Service Broker

Storage

Security

Server Objects

Replication

Management

XEvent Profiler

QLSinhVien...nhVD (102)*

DESKTOP-3QA... - Diagram_0

```
100 sv.DiaChi,
101 sv.DienThoai,
102 k.Ten AS TenKhoa
103 FROM SinhVien sv
104 INNER JOIN Khoa k
105 ON sv.MaKhoa = k.Ma
106 WHERE sv.MaKhoa = 'CNTT';
107
108 SELECT
109 k.Ma AS MaKhoa,
110 k.Ten AS TenKhoa,
111 COUNT(sv.Ma) AS SoLuongSinhVien
112 FROM Khoa k
113 LEFT JOIN SinhVien sv
114 ON k.Ma = sv.MaKhoa
115 GROUP BY
116 k.Ma,
117 k.Ten;
```

100 %

4 0

Live Query Statistics

Execution plan

	MaKhoa	TenKhoa	SoLuongSinhVien
1	CNTT	Cong nghe thong tin	4
2	MAIA	Ngon ngu Anh	3
3	QTKD	Quan tri kinh doanh	3

Rows	Executes	Short Text	Strid	NodeId	Parent	PhysicalOp	LogicalOp	Argument	DefinedValues	EstimateRows	EstimateIO	
1	3	1	SELECT	k.Ma AS MaKhoa, k.Ten AS TenKhoa	1	1	0	NULL	NULL	NULL	3	NULL
2	0	0	I-Compute Scalar	DEFINE ([Expr1002]=CASE WHEN [Expr1002] IS NULL T...	1	2	1	Compute Scalar	Compute Scalar	DEFINE ([Expr1002]=CASE WHEN [Expr1002] IS NULL THEN ([E...	3	0
3	3	1	I-Nested Loops	Left Outer Join, OUTER REFERE...	1	3	2	Nested Loops	Left Outer Join	OUTER REFERENCES ([I].[Ma])	3	0
4	3	1	I-Clustered Index Scan	OBJECT ([QLSinhVien_K240TVT002].[dbo].[SinhVie...	1	4	3	Clustered Index Scan	Clustered Index Scan	OBJECT ([QLSinhVien_K240TVT002].[dbo].[SinhVie...	3	0.003125
5	0	0	I-Compute Scalar	DEFINE ([Expr1002]=CONV...	1	5	3	Compute Scalar	Compute Scalar	DEFINE ([Expr1002]=CONVERT_IMPLICIT(int,[Expr1007],0)	1	0
6	3	3	I-Stream Aggregate	DEFINE ([Expr1007]=C...	1	6	5	Stream Aggregate	Aggregate	NULL	1	0
7	10	3	I-Clustered Index Scan	OBJECT ([QLSinhVien_K240TVT002].[dbo].[SinhVie] [P...	1	7	6	Clustered Index Scan	Clustered Index Scan	OBJECT ([QLSinhVien_K240TVT002].[dbo].[SinhVie] [P...	3.333333	0.0032035